

Member 1 Task Sheet

Lead Developer (FastAPI Backend, RAG Retriever & Database Core)

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1. Core Role Focus

You are responsible for writing the FastAPI application, setting up the relational database models using SQLAlchemy, creating the local legal database index, mapping RAG retrieval queries, and coding the contextual chatbot memory routers.

2. Directories to Create and Manage

- **backend/** Create database connectors and schemas.
- **backend/data/** Store local legal database JSON index.
- **backend/services/** Write RAG and session managers.
- **backend/routers/** Route REST endpoints.

3. Target File Checklists & Instructions

A. backend/requirements.txt (Day 1 - Highest Priority)

Add core libraries: fastapi, uvicorn, sqlalchemy, pymysql, python-dotenv, google-generativeai.

B. backend/database.py & models.py (Day 1 - Highest Priority)

Setup SQLAlchemy database connectors. Create tables User (id, preferred_language), Document (id, user_id, filename, doc_type, raw_text, summary_explanation, extracted_dates_json, legal_references_json, checklist_json), and ChatLog (id, document_id, role, content). Map fallback default engine to SQLite 'sqlite:///nyayamitra.db'.

C. backend/data/laws_kb.json (Day 2 - High Priority)

Compile key Indian laws (e.g. rent eviction rules, consumer defaults) into structured JSON file with plain-English translations.

D. backend/services/rag_retriever.py (Day 2 - High Priority)

Write text similarity matching comparing uploaded notice keywords to our local laws database. Perform searches in-memory (TF-IDF/cosine similarity) to bypass SQLite compilation blockers on Windows.

E. backend/routers/chat.py & services/chat_manager.py (Day 3 - Medium Priority)

Code chat endpoints. Store user questions and chatbot response cards inside chat_logs database table. Ensure session states reset when user uploads a new notice.

F. backend/services/calendar_generator.py (Day 4 - Low Priority)

Code calendar generator generating dynamic download schedule reminder invitation files (.ics).

4. Daily Schedule Checklist

- **Day 1:** Configure database schemas and test SQLite file connections locally.
- **Day 2:** Create JSON laws index and deploy the in-memory RAG indexer.
- **Day 3:** Integrate chatbot routes and write user messages directly to SQL logs.
- **Day 4:** Create calendar file generators and perform code validation checks.